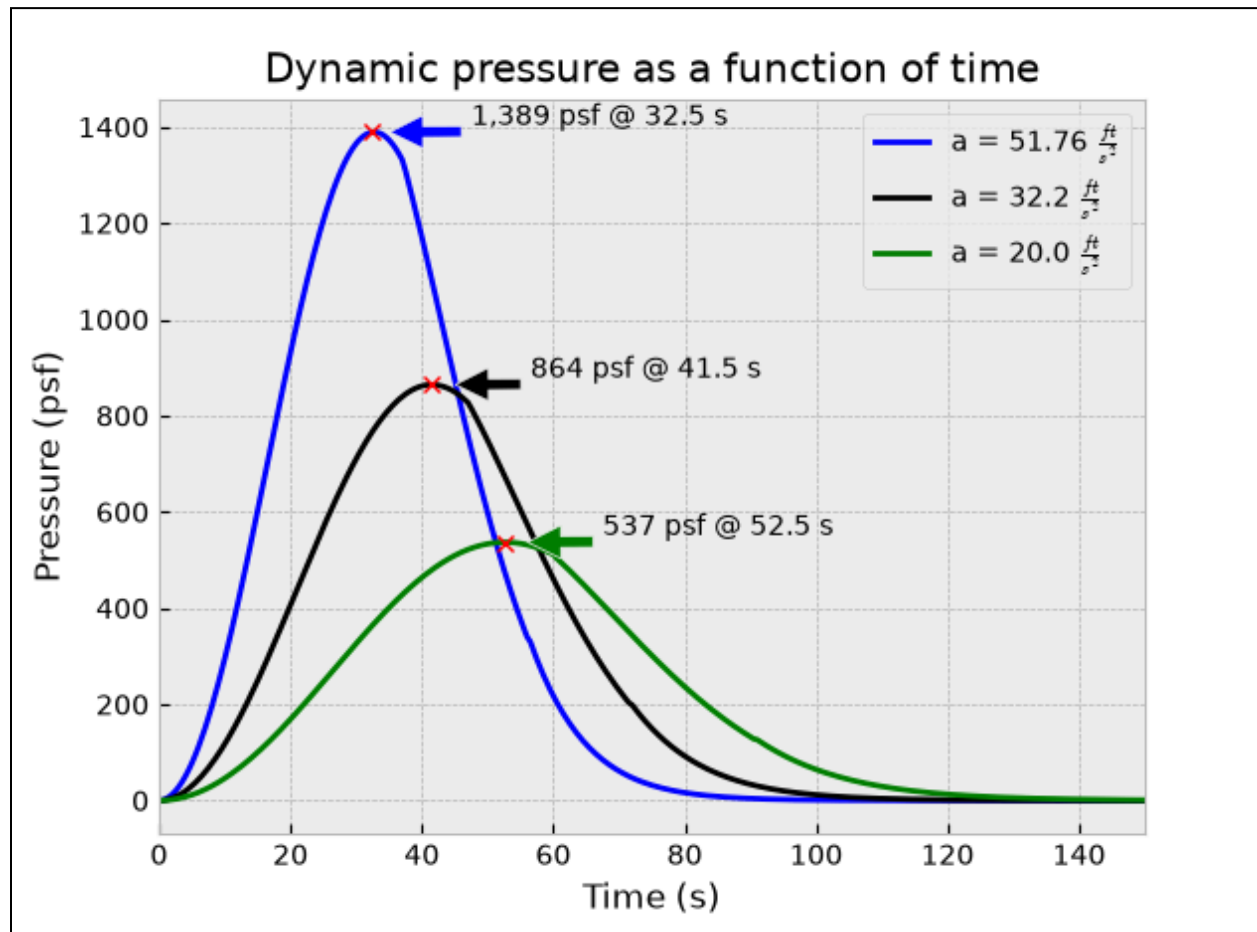


chapter 9 - pweave report



This Python script, formatted for pweave report generation, calculates and visualizes the dynamic pressure experienced by a rocket during launch under three distinct constant acceleration profiles. It uses basic kinematic equations to determine altitude and velocity over time, feeding that altitude into a segmented atmospheric model to accurately compute air density across different atmospheric layers. Finally, the script uses Matplotlib to graph the resulting pressure curves, automatically identifying and visually annotating the peak aerodynamic stress point (Max Q) for each simulated trajectory.